# Business Question and Visualization Report

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| Date | 29 October 2026 |
| Team ID | Not provided — confirm from MySkillWallet/college |
| Project Name | EV Market Intelligence Dashboard |
| Maximum Marks | 5 Marks |

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

**Business Questions and Visualisation**

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

**Sample**

1. **What are the monthly sales trends?**
   - *Code*
   - *Screenshot of visualisation*
2. **Which products have the highest revenue?**
   - *Code*
   - *Screenshot of visualisation*
3. **How is customer satisfaction varying across regions?**
   - *Code*
   - *Screenshot of visualisation*

**Note:** Min 8 business question and visualisations Required inform of above

# Project Context

This project analyzes the supplied EV market dataset (100,000 records, 37 columns) in Tableau Public. The objective is to turn vehicle, battery, sales, geography, customer-rating, range, and sustainability fields into nine views that support comparison and exploration.

# Business Questions and Visualisations

## 1. How do annual sales units compare across vehicle categories?

Visualisation: Horizontal bar chart: Category on Rows; SUM(Annual Sales Units) on Columns; sort descending.

Finding from the supplied CSV: Identifies categories with the largest reported sales volume. The dataset's highest total is Two-Wheeler Scooter (85,031,983); Electric Bus is lowest (5,323,646).

Screenshot: Insert the corresponding Tableau worksheet/dashboard screenshot here before submission.

## 2. Which 10 manufacturers have the highest annual sales units?

Visualisation: Horizontal bar chart: Manufacturer and SUM(Annual Sales Units); sort descending; filter Top 10 by field.

Finding from the supplied CSV: SurgeWorks leads the displayed top 10 at 4,883,639 annual sales units, followed by CurrentMobility at 4,779,221.

Screenshot: Insert the corresponding Tableau worksheet/dashboard screenshot here before submission.

## 3. What is the distribution of battery technologies in the dataset?

Visualisation: Pie chart: Battery Type on Color; COUNT(Vehicle Id) on Angle; show labels/percentages.

Finding from the supplied CSV: NMC is the most common battery type (33,836 records), followed by LFP (32,101). This is a record mix, not sales share.

Screenshot: Insert the corresponding Tableau worksheet/dashboard screenshot here before submission.

## 4. How does average driving range vary by vehicle category?

Visualisation: Bar chart: Category and AVG(Range Km); sort descending.

Finding from the supplied CSV: Luxury Sedan has the highest average range (~549 km), while Two-Wheeler Scooter has the lowest (~105 km).

Screenshot: Insert the corresponding Tableau worksheet/dashboard screenshot here before submission.

## 5. Which manufacturing countries are represented, and how are records distributed geographically?

Visualisation: Map: Manufacturing Country mapped to geographic role; use COUNT(Vehicle Id) or SUM(Annual Sales Units) for size; tooltip includes country.

Finding from the supplied CSV: Shows geographic spread of records. Use the map as a distribution view; do not infer production volumes beyond the selected measure.

Screenshot: Insert the corresponding Tableau worksheet/dashboard screenshot here before submission.

## 6. Which vehicle categories have the highest average customer rating?

Visualisation: Bar chart: Category and AVG(Customer Rating); sort descending; optionally exclude null ratings.

Finding from the supplied CSV: Ratings are tightly clustered. Crossover averages ~4.10, while Electric Truck averages ~4.07 among non-null values.

Screenshot: Insert the corresponding Tableau worksheet/dashboard screenshot here before submission.

## 7. How have annual sales units varied by model year?

Visualisation: Line chart: Model Year on Columns; SUM(Annual Sales Units) on Rows.

Finding from the supplied CSV: Sales totals vary by year. 2021 is highest in the provided data (~88.09M); 2025 is lowest (~82.55M).

Screenshot: Insert the corresponding Tableau worksheet/dashboard screenshot here before submission.

## 8. What is the relationship between battery capacity and driving range?

Visualisation: Scatter plot: Battery Capacity KWh on Columns; Range Km on Rows; color by Battery Type; each mark represents a vehicle/model record.

Finding from the supplied CSV: The plot helps examine the broad association and outliers; it does not prove battery capacity alone causes longer range.

Screenshot: Insert the corresponding Tableau worksheet/dashboard screenshot here before submission.

## 9. Which vehicle categories report the greatest annual $CO_2$ savings?

Visualisation: Bar chart: Category and AVG(CO2 Saved Tons Per Year); sort descending.

Finding from the supplied CSV: Electric Bus, Pickup Truck, and Electric Truck have the highest average reported savings (~9.05, ~9.01, and ~8.96 tons/year respectively). Verify assumptions behind this synthetic dataset metric before using it as a real-world claim.

Screenshot: Insert the corresponding Tableau worksheet/dashboard screenshot here before submission.

*Note: Findings are computed from the supplied CSV, not from an external market source. Dashboard screenshots and the public Tableau URL must be inserted after final verification.*